

EASTERN INTERNATIONAL UNIVERSITY



MIS 443

Business Data Management

Assignment 4

Individual PostgreSQL Database Project

Lecturer: Dang Thai Doan

Prepared by

Major: Business Analytics

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Quarter 4/2025-2026

1. Introduction to the School database:

The School Database is a relational database management system designed to support academic and administrative operations within a higher education institution. The primary purpose of the system is to centralize and efficiently manage core educational entities, including students, faculty members (professors), academic courses, and course enrollments.

By structuring data into a normalized relational schema, the database enables various departments- such as the Registrar's Office, Academic Departments, Financial Aid, and Curriculum Committees- to retrieve key operational insights, perform audits, monitor academic performance, and streamline course administration.

2. Description of the main tables and relationships

2.1 **sch.students**: Stores demographic and academic status information of enrolled students.

Primary Key: student_id (Integer) uniquely identifies each student

Attributes: first_name, last_name, email, enrollment_date, graduation_year, and major.

2.2 **sch.professors**: Manages information about faculty members.

Primary Key: professor_id (Integer) uniquely identifies each professor.

Attributes: first_name, last_name, email, department, and hire_date.

2.3 **sch.courses**: Contains information on academic courses offered by various departments.

Primary Key: course_id (VARCHAR(10)) uniquely identifies each course code (e.g., CS101, MATH201).

Foreign Key: professor_id references sch.professors(professor_id) to specify which faculty member teaches the course.

Attributes: course_name, credits, department

2.4 **sch.enrollments**: Serves as a junction/bridge table linking students to the courses they are enrolled in, establishing a **Many-to-Many relationship** between students and courses.

Primary Key: enrollment_id (Integer) uniquely identifies each enrollment transaction.

Foreign Keys:

- student_id references sch.students(student_id).
- course_id references sch.courses(course_id).

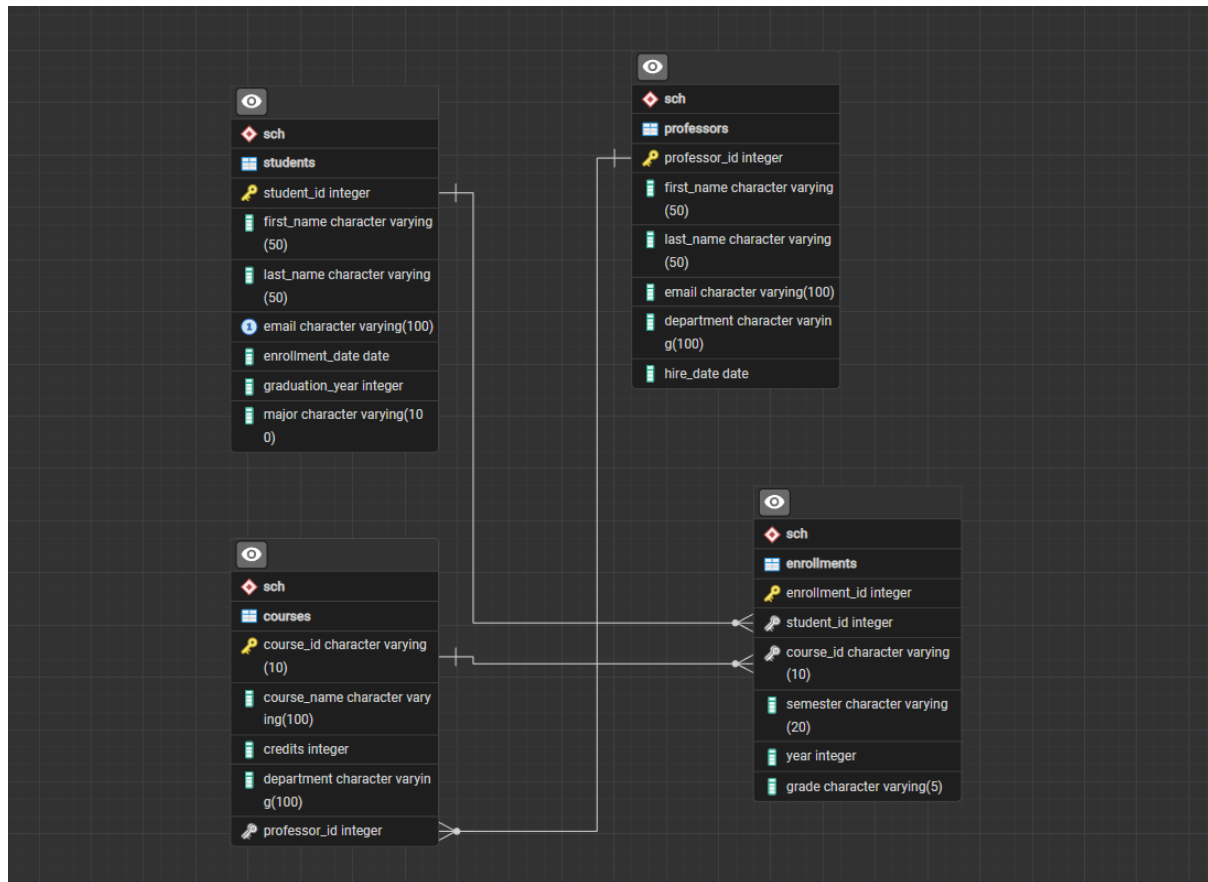
Attributes: semester, year, and grade

Relationships Summary:

- **Professors to courses: 1-to-Many (1:N)** - One professor can teach multiple courses, but each course is assigned to one primary professor.
- **Students to enrollments: 1-to-Many (1:N)** - One student can have multiple enrollment records.

- **Courses to enrollments: 1-to-Many (1:N)** - One course can have multiple student enrollment records

3. Entity–Relationship Diagram



4. Summary of the database implementation

The database implementation was conducted in PostgreSQL 16 using pgAdmin 4 as the management and execution platform. Key steps included:

- **Schema Creation**: Created a custom schema sch to group related educational tables logically.
- **Table Definition & Constraints**: Structured tables with appropriate data types (e.g., VARCHAR, INTEGER, DATE) and defined explicit Primary Key and Foreign Key constraints to maintain referential integrity across related entities.
- **Data Ingestion**: Populated tables using standard SQL INSERT INTO statements and CSV data imports for students, professors, courses, and enrollments.
- **Query Execution**: Successfully drafted and executed data manipulation queries (DML) to answer administrative and academic business requirements.

5. Summary of the completed SQL questions

Below is the summary of SQL queries executed during the assignment:

Question 6.1: Full student directory audit.

- **Query**: `SELECT * FROM sch.students;`
- **Result**: Successfully retrieved all 7 student records.

Question 6.2: List of enrolled Computer Science students for orientation.

- **Query:** SELECT first_name, last_name, graduation_year FROM sch.students WHERE major = 'Computer Science';
- **Result:** Returned 2 students (Alex Thompson, Emily Chen).

Question 6.3: Course catalog ranked by academic workload.

- **Query:** SELECT course_name, credits FROM sch.courses ORDER BY credits ASC; *(Note: or DESC for highest to lowest).*
- **Result:** Listed courses ordered by credit weight.

Question 6.: Expected graduates for the Class of 2026.

Query: SELECT first_name, last_name, major FROM sch.students WHERE graduation_year = 2026;

- **Result:** Returned 3 expected graduates (Alex Thompson, Maria Rodriguez, Robert Johnson).

Question 6.5: Total count of active course offerings.

- **Query:** SELECT COUNT(course_id) AS total_courses FROM sch.courses;
- **Result:** Total of 6 available courses.

Question 6.6: Average credit load benchmark per course.

- **Query:** SELECT AVG(credits) AS average_credits FROM sch.courses;
- **Result:** Calculated average course credits (nearly 3.67 credits per course).

6. Selected pgAdmin screenshots

6.1. The registrar is auditing the full student directory ahead of the new academic term. Return the complete student roster from the students table.

Query

```

1  ---1. The registrar is auditing the full student directory ahead of the new academic term.
2  ---Return the complete student roster from the students table.
3  select *from sch.students;
4
5  ---2. The Computer Science department needs a list of all enrolled CS students for an upcoming orienta
6  ---Return students who are majoring in Computer Science.
7

```

Data Output

student_id	first_name	last_name	email	enrollment_date	graduation_year	major
1001	Alex	Thompson	alex.thompson@university.edu	2022-09-01	2026	Computer Science
1002	Maria	Rodriguez	maria.rodriguez@university.e...	2022-09-01	2026	Mathematics
1003	James	Wilson	james.wilson@university.edu	2021-09-01	2025	Physics
1004	Emily	Chen	emily.chen@university.edu	2023-09-01	2027	Computer Science
1005	Robert	Johnson	robert.johnson@university.edu	2022-09-01	2026	Chemistry
1006	Sarah	Davis	sarah.davis@university.edu	2021-09-01	2025	Biology
1007	David	Kim	david.kim@university.edu	2024-09-01	2028	NULL

Total rows: 7 Query complete 00:00:00.144

6.2. The Computer Science department needs a list of all enrolled CS students for an upcoming orientation session.
Return students who are majoring in Computer Science.

Query

```

1  ---1. The registrar is auditing the full student directory ahead of the new academic term.
2  ---Return the complete student roster from the students table.
3  select *from sch.students;
4
5  ---2. The Computer Science department needs a list of all enrolled CS students for an upcoming orienta
6  ---Return students who are majoring in Computer Science.
7  select first_name, last_name, graduation_year
8  from sch.students
9  where major = 'Computer Science';

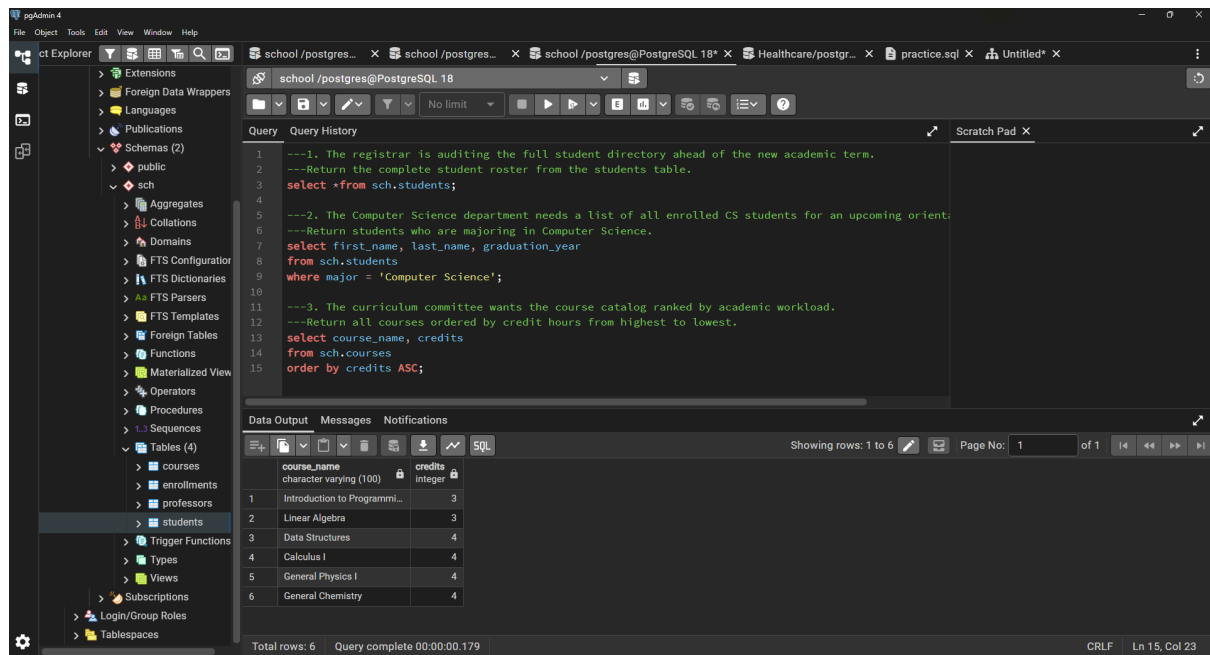
```

Data Output

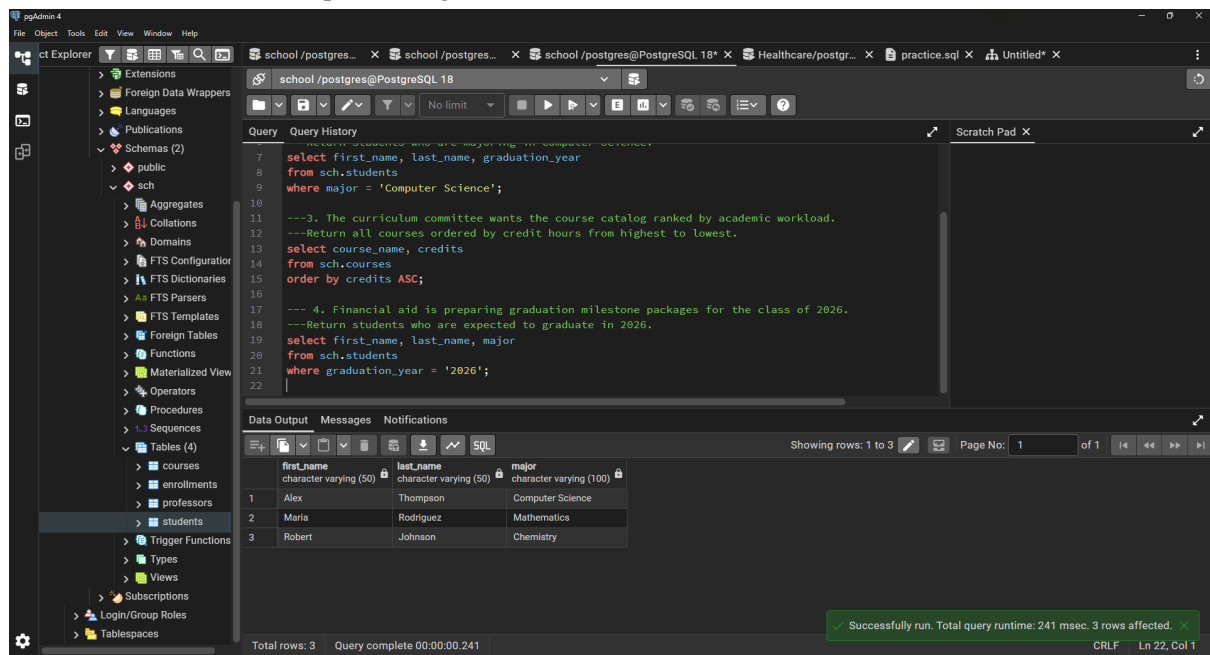
first_name	last_name	graduation_year
Alex	Thompson	2026
Emily	Chen	2027

Total rows: 2 Query complete 00:00:00.140

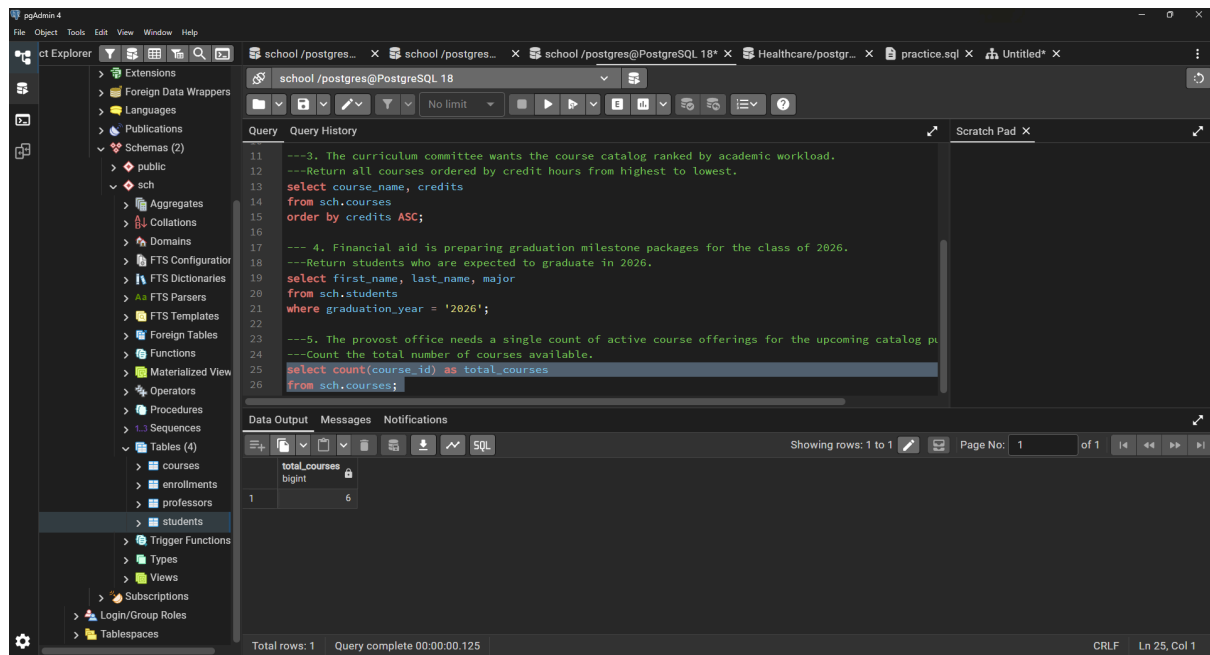
6.3. The curriculum committee wants the course catalog ranked by academic workload.
Return all courses ordered by credit hours from highest to lowest.



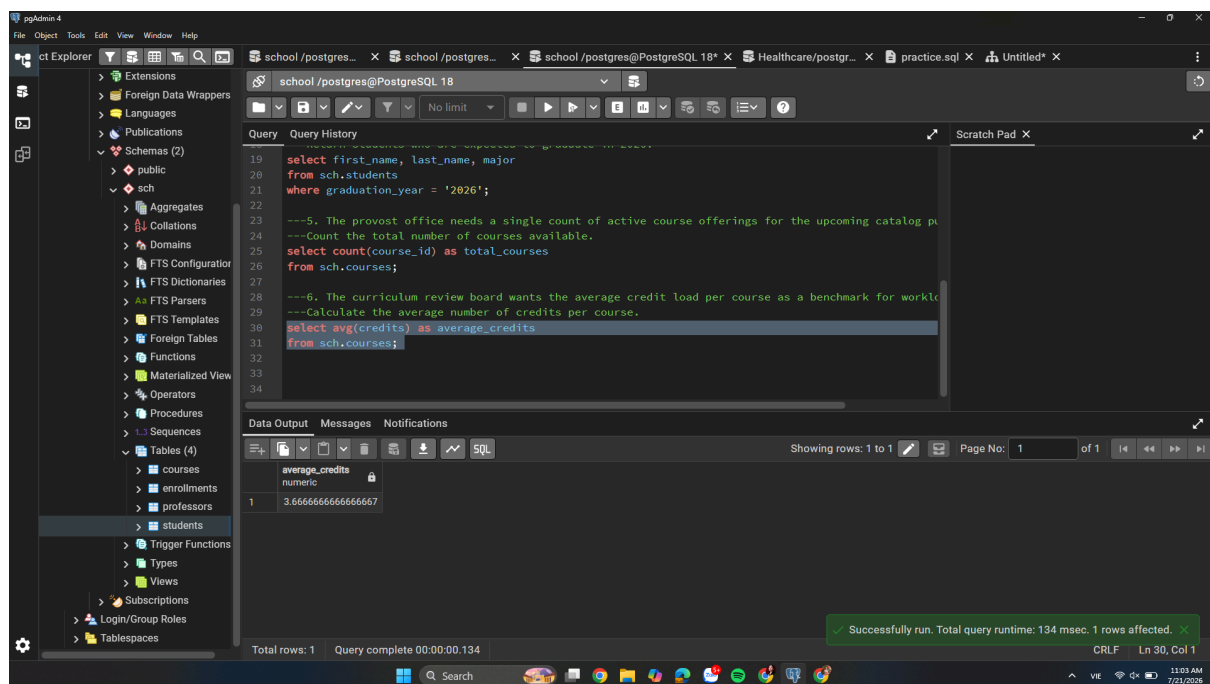
6.4 Financial aid is preparing graduation milestone packages for the class of 2026.
Return students who are expected to graduate in 2026.



6.5 The provost office needs a single count of active course offerings for the upcoming catalog publication.
Count the total number of courses available.



6.6. The curriculum review board wants the average credit load per course as a benchmark for workload balance.



7. Challenges encountered and solutions:

Data Type Conflict during CSV Data Import

Problem: When importing data from CSV files into PostgreSQL tables using pgAdmin's Import/Export tool, data type conflicts occurred (e.g., mismatch between numeric/date formatting in CSV files and strict column data types like INTEGER or DATE in PostgreSQL), causing import process failures.

Solution: Adjusted column data types during table creation to match data formats accurately, cleaned input CSV data prior to ingestion, and ensured schema definitions strictly aligned with the incoming row values before executing the import.

8. Conclusion:

The School Database Project successfully demonstrated the end-to-end implementation of a relational database system using PostgreSQL 16 and pgAdmin 4. By overcoming real-world data integration challenges- such as resolving CSV data type import conflicts and maintaining strict referential integrity- the database established a solid schema foundation under the sch namespace.

The completed project proves the capability of relational databases to centralize university operations, streamline administrative reporting, and answer business queries for key academic departments (such as the Registrar, Financial Aid, and Provost offices). Ultimately, this assignment strengthened practical expertise in data modeling, SQL query writing, schema normalization, and data troubleshooting in a PostgreSQL environment.

9. Personal GitHub repository link

<https://github.com/NhuqDang/MIS-443---Business-Data-Management>